	diagnostics, measure performance, validate data transfers, and may allow viewing of SMART information or logs outside a host system. These are primarily used by repair and QA environments to quickly screen or refurbish drives.

Note: this document is not about design testing. Rather about testing to identify if no longer meets design specification.

62. FAULT FINDING AND TROUBLESHOOTING

Effective troubleshooting requires a structured approach that addresses common failure modes, leverages built-in diagnostics, and uses available error logs and reporting data.

The core troubleshooting workflow is:

1. Verify connections and power. Look for damaged connectors or bent pins. Check that the data and power cables are firmly connected; try a different cable or connection point.
2. Confirm that the host system recognizes the device. If not detected, test the storage device in a tester, or a known working system to isolate the issue. For RAID or network-attached storage, ensure controllers & network connections are healthy.
3. Review system and device logs. Look for storage errors, disconnections, or timeout messages; reallocated/bad sectors, pending or uncorrectable sector counts, sudden increases in read/write errors, temperature or power cycle anomalies.
4. Run diagnostic tests. On complex controllers, run array and member disk diagnostics, check for degraded or rebuilding states.
5. Assess performance and symptoms. Slow performance or pauses can indicate physical issues, excessive write amplification, or pending media failure. Frequent file-system checks or“drive needs to be repaired” warnings indicate pending media or controller failure.
6. Check for environmental issues. Overheating, vibration, shock, or high humidity can accelerate storage failures. Review environmental sensor and exposure logs if available.
7. Isolate and replace if needed. If diagnostics point to device-level failure, replace the drive, back up salvageable data first, if possible. For intermittent problems, swap cables/ports and repeat tests to identify the faulty link. In RAID, follow the controller’s procedure for replacing and rebuilding failed drives.
8. Restore and Monitor. After repair or replacement, restore data from backups if necessary. Continue monitoring SMART and error statistics for early signs of recurrence.

62.1. RAM

For triage of no-boot, training failure, or ECC storms, fall back to JEDEC baseline first. Confirm SPD readability. Retest with a known-good module if instability persists.

After DIMM replacement, confirm the new module’s SPD is readable and that the intended profile (JEDEC vs XMP/EXPO profile index) matches platform policy. Mixed kits or a module without the expected profile fall back to JEDEC or fail training.